

PROJECT TOPIC: AGRICULTURAL CROP LOSS DUE TO INSECT PESTS & DISEASES

Topic Description: Field Survey of Local Agricultural Farms Documenting Crop Losses Inflicted by Insect Pests and Fungal Pathogens in Recent Years, Monetary Damages Incurred, and Farmer Perspectives on Causation

Agriculture is the bedrock of Maharashtra's rural economy, providing livelihoods for millions of smallholder farming families across diverse agro-climatic zones. However, crop cultivation in recent years has become an increasingly high-risk, economically volatile enterprise. Farmers across intensive agricultural belts—such as Nashik, Pune, Ahmednagar, Kolhapur, and Marathwada—are grappling with devastating pest outbreaks and recurrent crop disease epidemics that destroy standing crops on an unprecedented scale.

While insect pests and plant pathogens have always existed within natural ecosystems, their destructiveness has escalated drastically in recent years. Indiscriminate, excessive application of broad-spectrum synthetic chemical pesticides has annihilated natural biological predators (such as ladybird beetles, spiders, and parasitic wasps), inducing widespread **Pesticide Resistance** and triggering explosive secondary pest resurgence. Concurrently, erratic climatic anomalies—such as prolonged unseasonal humidity, sudden spring heatwaves, and delayed winters—have disrupted traditional biological diapause cycles, creating continuous, overlapping breeding generations of invasive and native insect pests.

This project conducts a comprehensive, on-the-ground agro-ecological field survey across 5 operational farms in our regional agricultural periphery. The study documents the specific insect pests and fungal diseases afflicting major cash and food crops (Soybean, Cotton, Tomato, ¹Pomegranate, Onion), quantifies the devastating monetary losses incurred by farmers, investigates farmer perspectives

2. Importance of the topic (विषयाचे महत्त्व)

Investigating agricultural pest damage is vital for regional food security, farm economics, and environmental toxicology:

Critical Dimensions of Agricultural Pest Loss Assessment :-

- **Preventing Catastrophic Agrarian Debt & Farmer Distress:** Heavy pest devastation wipes out farmers' seasonal income while leaving them burdened with massive debts incurred from purchasing expensive hybrid seeds, chemical fertilizers, and proprietary pesticides.
- **Breaking the Vicious 'Pesticide Treadmill':** When pests develop physiological resistance to standard chemical sprays, farmers respond by spraying higher doses and more toxic chemical cocktails, accelerating environmental toxicity without controlling pests.
- **Preserving Native Agro-Biodiversity:** Excessive chemical spraying decimates beneficial pollinator populations (honeybees and solitary bees), causing severe pollination failure in neighboring fruit orchards and cross-pollinated vegetable crops.
- **Eliminating Toxic Chemical Residues in the Food Chain:** Desperate farmers frequently spray toxic systemic insecticides right up to the day of harvest, leaving dangerous pesticide residues on commercial vegetables that enter urban food markets.

Understanding the root ecological causes of pest outbreaks is the indispensable first step toward sustainable agriculture.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This agricultural pest loss investigation was structured around structured empirical, economic, and agronomic objectives:

Structured Project Objectives :-

- **Farm Field Survey & Farmer Interviews:** To visit 5 diverse operational agricultural farms cultivating commercial food and cash crops in our local agrarian block.
- **Pest & Disease Identification Matrix:** To identify and catalog the primary insect pests (sucking, boring, defoliating) and fungal/bacterial diseases affecting standing crops in recent seasons.
- **Quantifying Monetary & Yield Losses:** To record the percentage of crop yield destroyed and calculate the net monetary financial loss (in Rupees per acre) incurred by each farmer.
- **Farmer Perspectives on Root Causes:** To document farmers' personal views regarding why pest outbreaks have intensified (e.g., climate change, fake pesticides, monoculture, soil degradation).
- **Integrated Pest Management (IPM) Framework:** To synthesize practical biological, cultural, and mechanical pest management strategies to eliminate toxic chemical dependency.

The study provides grounded empirical insights into the frontline challenges of agrarian survival.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A structured field inspection, crop loss accounting, and semi-structured farmer interview methodology was deployed:

Methodological Phases Executed :-

- **Phase 1 – Farm Site & Crop Selection:** Selected 5 representative commercial farms cultivating 5 dominant regional crops: Farm 1 (Soybean), Farm 2 (Cotton), Farm 3 (Tomato), Farm 4 (Pomegranate), and Farm 5 (Onion).
- **Phase 2 – Direct Farmer Questionnaire Administration:** Conducted in-depth, on-farm interviews with each farmer covering crop acreage, pest infestation history over the last 3 years, chemical spraying schedules, and input costs.
- **Phase 3 – Physical Field Damage Inspection:** Inspected standing crop plots to photograph and document characteristic pest symptoms (leaf curling, fruit boreholes, stem galling, fungal blights).
- **Phase 4 – Economic Loss Calculation:** Computed net monetary losses utilizing the formula: $\text{Monetary Loss} = (\text{Normal Expected Yield} - \text{Actual Harvested Yield}) \times \text{Market Price} + \text{Wasted Pesticide Spray Costs}$.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Humans and Environment: Agriculture) and Chapter 3 (Sustainable Agriculture) of the HSC Standard 12 EVS curriculum.

All recorded yield figures and financial losses reflect authenticated farm ledger data and direct farmer testimonies.

5. Observations (निरीक्षणे)

Field survey of 5 agricultural farms in the local agrarian belt revealed devastating pest infestations and crushing financial losses incurred in recent seasons:

5.1 Agricultural Crop Loss, Pest Disease Profile & Monetary Damage Matrix

Crop & Acreage	Primary Pest / Disease Identified	Yield Loss Percentage	Net Monetary Loss (Per Acre)	Primary Root Cause Cited by Farmer
Soybean (3.5 Acres)	Spodoptera litura (Tobacco Caterpillar) & Girdle Beetle; severe larval defoliation of leaves.	45% crop loss (Harvest dropped from 10 quintals to 5.5 quintals/acre).	₹18,500 / acre (Lost yield: ₹15,750 + ₹2,750 in ineffective sprays).	Prolonged August dry spell followed by sudden humid cloudy days; complete failure of generic chemical sprays.
Bt Cotton (4.0 Acres)	Pink Bollworm (Pectinophora gossypiella); larvae bore into developing bolls, destroying lint.	50% crop loss (Bolls failed to open; stained, unmarketable fiber).	₹28,000 / acre (Crushing loss on expensive hybrid inputs).	Pest developed complete genetic resistance to Cry toxins in Bt cotton; continuous mono-cropping without refuge crops.
Tomato (2.0 Acres)	Tomato Leaf Curl Virus (ToLCV)	65% crop loss (Stunted yellow	₹42,000 / acre (Intensive drip	Extreme spring heatwaves triggered explosive

6. Analysis of data (माहितीचे विश्लेषण)

Agronomic and ecological analysis of the survey findings exposes the severe biological collapse induced by chemical-centric monoculture:

Core Analytical Inferences :-

- **The 'Pesticide Treadmill' and Resistance Evolution:** The catastrophic failure of Bt cotton against Pink Bollworm illustrates basic evolutionary selection. Farmers abandoned the mandatory planting of non-Bt 'refuge' border rows. Consequently, the rare individual insect possessing natural resistance genes mated with other resistant insects, creating a 100% resistant pest population within a decade. Spraying heavy synthetic pyrethroids merely killed off beneficial spider predators, leaving resistant bollworms entirely unchecked.
- **Chemical Nitrogen Fertilizer Induced Susceptibility:** Analysis of onion and tomato management revealed that farmers applied nearly double the recommended doses of chemical Urea to force rapid vegetative growth. High nitrogen makes plant tissue extraordinarily tender, succulent, and high in free amino acids, effectively creating an irresistible biological magnet for sap-sucking thrips, aphids, and whiteflies.
- **Climate Volatility as an Entomological Catalyst:** The devastating bacterial blight (Telya) in pomegranate was directly triggered by unseasonal April hailstorms. Hailstones inflicted microscopic lesions on fruit rinds, allowing airborne *Xanthomonas* bacteria to penetrate deep into fruit pulp during subsequent warm, humid days. Climate destabilization is fundamentally altering pest epidemiology.
- **The Necessity of Ecological Pest Management:** Chemical poisons cannot outrun biological evolution. Sustainable pest suppression requires restoring ecological balance through multi-cropping, pheromone mating-disruption traps, botanical repellents, and biological parasitoids.

The analysis proves that agricultural 6 epidemics are symptoms of severe agro-ecosystem imbalance.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Surveying local farms proves that insect pests and diseases are inflicting catastrophic, debt-inducing monetary losses upon Maharashtra's farming communities. Chemical pesticide dependency has failed due to pest resistance and the destruction of natural predators. To safeguard farmer livelihoods and food security, agriculture must urgently transition to ecologically sound Integrated Pest Management (IPM).

Action Checklist for Integrated Pest Management (IPM)

Stewardship :-

- ❑ Install yellow sticky traps (against whiteflies/thrips) and light traps (against night-flying moths) across all fields.
- ❑ Deploy species-specific Pheromone Traps (5 per acre) to monitor and disrupt Pink Bollworm mating cycles.
- ❑ Plant flowering border rows of Marigold, Mustard, and Coriander to serve as natural trap crops and attract predator wasps.
- ❑ Spray biological entomopathogenic fungi (*Beauveria bassiana* and *Verticillium lecanii*) at initial pest emergence.
- ❑ Strictly eliminate prophylactic synthetic chemical spraying; spray only when pest counts exceed Economic Injury Levels.

8. Bibliography / References (संदर्भ सूची)

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